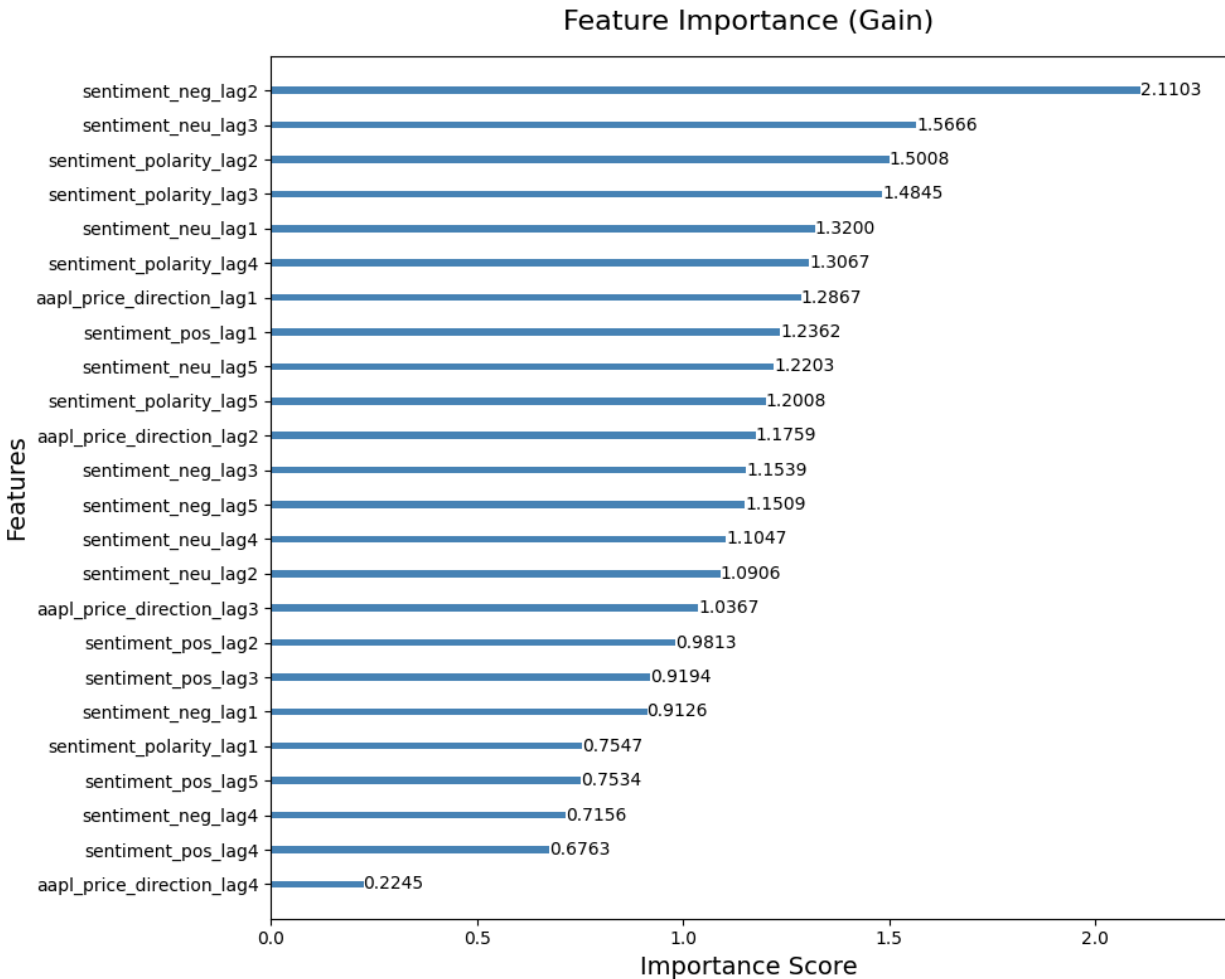


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The following chart ranks predictors in the classification model by order of importance (measured by gain in model loss):



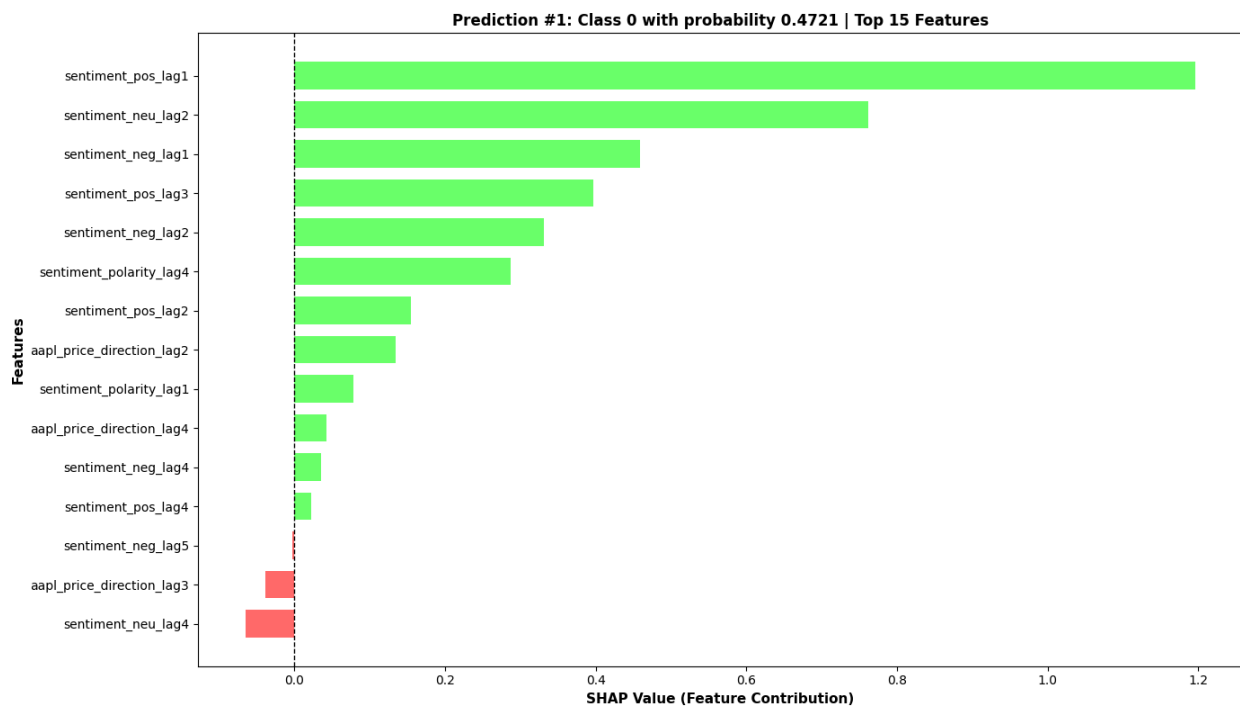
Overall, this is very interesting because the news sentiment scores ranked much higher than the historical price direction lags. This is perhaps not too surprising since the models trained on solely news sentiment data performed much better than the models trained solely on the historical price data in terms of accuracy and F1 scores. Of the historical price lags, the most recent direction lag was ranked highest, showing that recent price data is still most effective in predicting next-day price direction.

Within the sentiment data, the 2 and 3 day lags between the negative, neutral, and polarity scores were all ranked high in importance. This information suggests that it takes a couple days for news articles to impact stock price direction, and articles containing an extreme amount of negative language as well as extreme overall sentiment (measured through polarity) are correlated with a consumer's decision to buy or sell stock. Noticeably absent is the percentage of positive language, this ranked very low in feature importance.

Evaluation of 5 Random Predictions

The SHAP (SHapley Additive exPlanations) package is a popular Python library used to explain the output of any machine learning model. It is based on Shapley values from cooperative game theory, which provide a mathematically grounded way to fairly distribute "credit" among the features that contributed to a model's prediction.

The SHAP analysis helps explain how the model arrived at each prediction by showing which lagged sentiment features pushed the prediction toward Up (Class 1) or Down (Class 0). Across the five randomly selected cases, the model relied much more on recent sentiment patterns than on lagged price-direction variables. The most influential features overall were sentiment_pos_lag1, sentiment_neg_lag2, sentiment_neu_lag5, sentiment_pos_lag4, and sentiment_polarity_lag5. This suggests the model is mainly learning from the recent history of positive, negative, and neutral sentiment, especially within the prior one to five days. In these examples, four of the five predictions were correct, giving an 80% accuracy on this small sample, but the SHAP plots also show that the prediction logic is often driven by a few dominant sentiment features rather than a broad balance of all inputs.

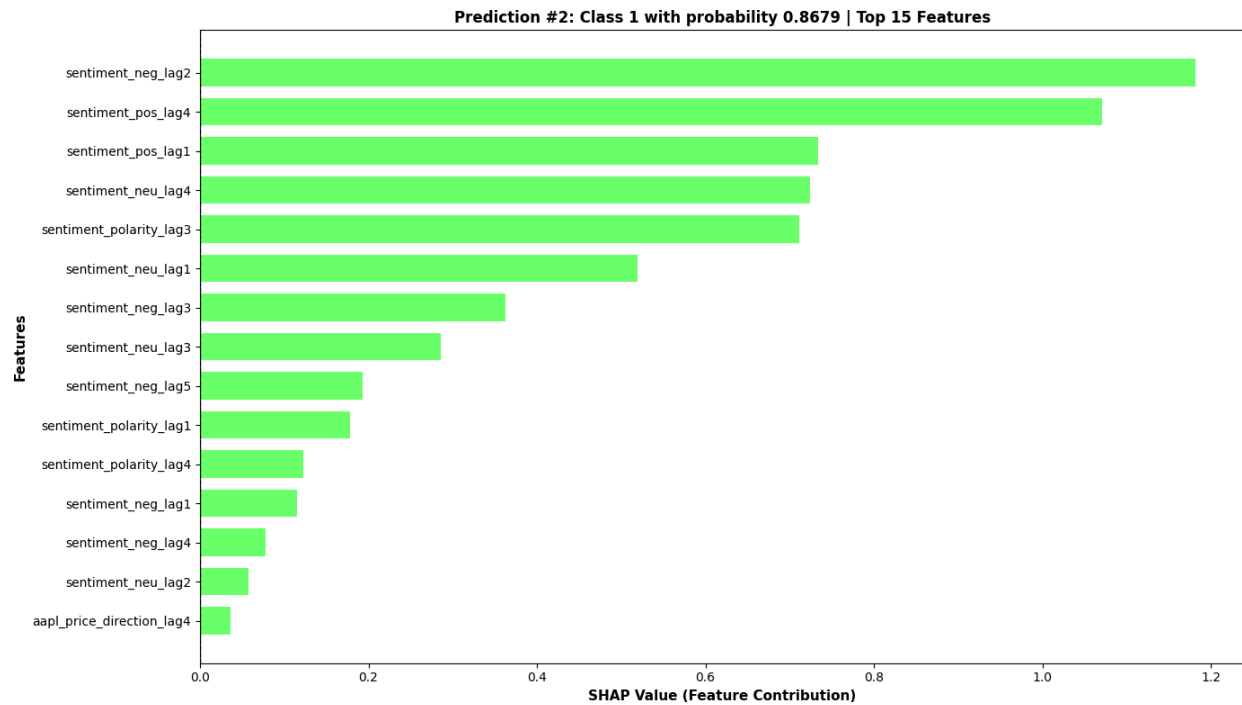


PREDICTION #1 GRAPH

For Prediction #1, the model predicted Down with probability 0.4721, although the true class was Up. This was a borderline prediction, which means the push toward Down was not overwhelming. The strongest features pushing the model toward Up were sentiment_pos_lag1 with a SHAP value of +1.1963 and sentiment_neu_lag2 with +0.7620, while the strongest features pulling it back toward Down were sentiment_neu_lag3 at -0.6583, sentiment_polarity_lag5 at -0.5501, and sentiment_neg_lag3 at -0.4840. In practical terms, the model saw strong recent positive sentiment, but it also detected earlier neutral and

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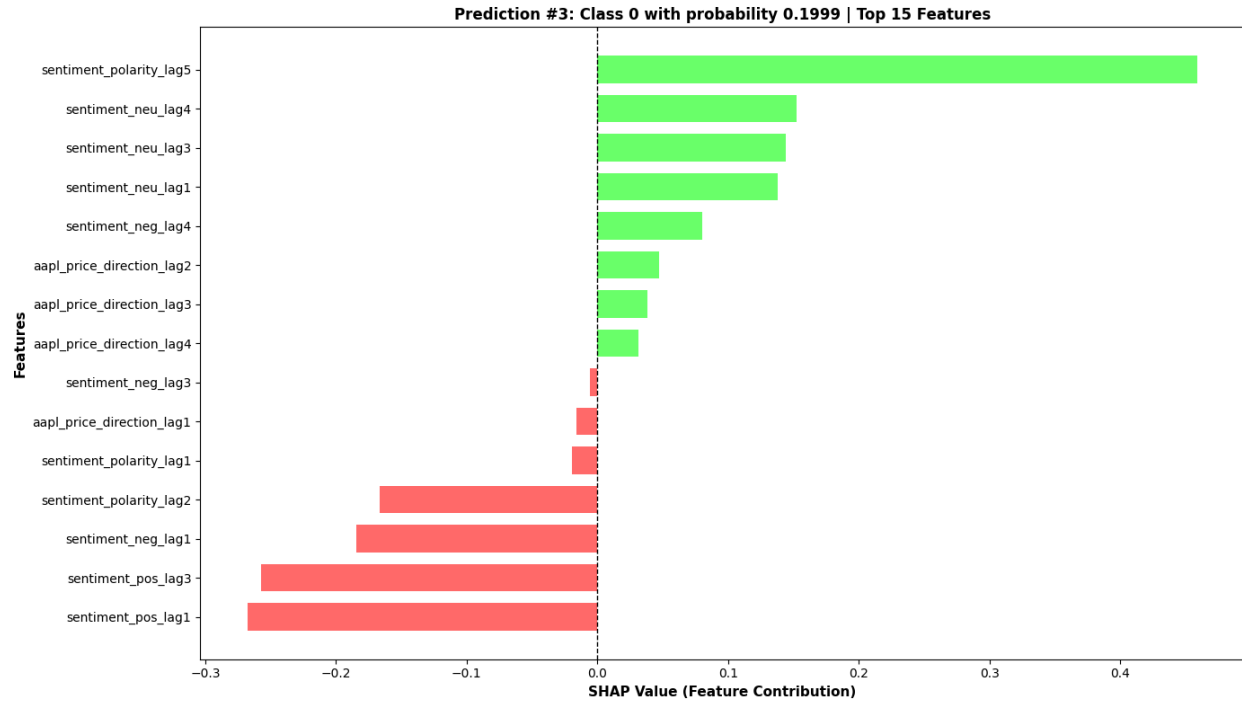
polarity patterns that it interpreted as bearish enough to cancel much of that positive signal. To significantly move this prediction and flip it to Up, the most useful changes would be to increase the features that currently push toward Down but could be shifted into a more bullish range. For example, sentiment_polarity_lag5 would need to rise from 0.4991 to about 0.6097, a change of 0.1106, and sentiment_polarity_lag3 would need to rise from 0.4438 to about 0.6022, a change of 0.1584. Smaller but still meaningful changes include increasing sentiment_neg_lag3 from 0.0285 to 0.0474 and sentiment_neu_lag1 from 0.8289 to 0.8794. Because the current probability is close to the decision boundary, moderate feature movement could plausibly reverse the final class.



PREDICTION #2 GRAPH

For Prediction #2, the model predicted Up with a high probability of 0.8679, and this prediction was correct. This was one of the clearest bullish decisions. The most important features pushing the model upward were sentiment_neg_lag2 (+1.1813), sentiment_pos_lag4 (+1.0713), sentiment_pos_lag1 (+0.7332), sentiment_neu_lag4 (+0.7243), and sentiment_polarity_lag3 (+0.7117). Although a few features such as sentiment_pos_lag3 and sentiment_polarity_lag5 pushed toward Down, their influence was much smaller than the positive contributors. To flip this prediction from Up to Down, the most important features would need to move in the opposite direction. For example, sentiment_pos_lag1 would need to fall from 0.1123 to about 0.0826, a decrease of 0.0297, while sentiment_pos_lag4 would need to drop by about 0.0109 and sentiment_polarity_lag3 would need to fall substantially from 0.5723 to 0.3820, a decrease of 0.1903. sentiment_neu_lag4 would also need to decline by 0.0267. Since the model was very confident here, flipping the class would likely require simultaneous movement in several of these top features rather than just one small adjustment.

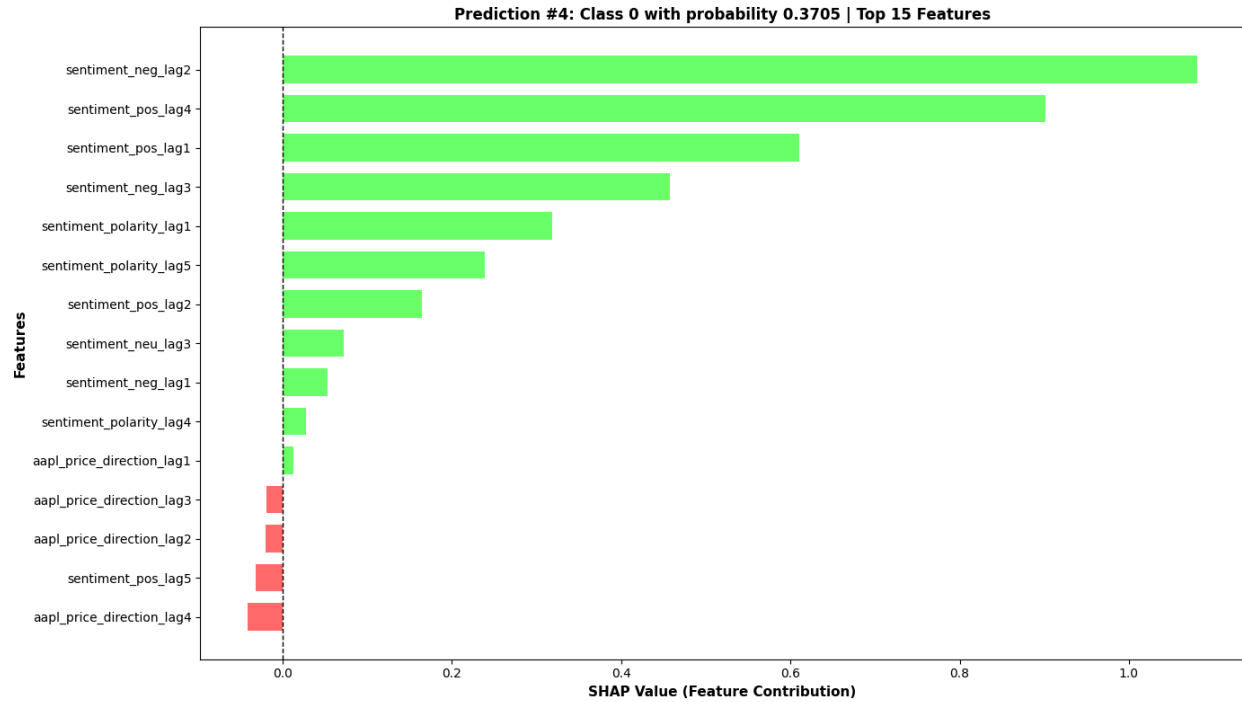
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PREDICTION #3 GRAPH

For Prediction #3, the model predicted Down with probability 0.1999, and this was also correct. This was a strong bearish prediction. The largest downward pressures came from sentiment_neu_lag5 (-1.1001), sentiment_neg_lag2 (-0.8143), sentiment_neu_lag2 (-0.4112), sentiment_polarity_lag3 (-0.3718), and sentiment_pos_lag2 (-0.3528). Only one major feature, sentiment_polarity_lag5, pushed meaningfully toward Up with +0.4591, but it was not large enough to outweigh the bearish signals. To move this prediction significantly and flip it to Up, the strongest targets would be the largest negative contributors. For example, sentiment_neu_lag5 would need to increase from 0.8455 to around 0.8792, a change of 0.0337, and sentiment_neu_lag2 would need to increase by 0.0328. sentiment_neg_lag2 would need to rise from 0.0393 to 0.0472, a smaller change of 0.0079, but because its SHAP value is large, even that small movement could matter. This prediction appears more stable than Prediction #1 because the negative contributions were both stronger and more concentrated.

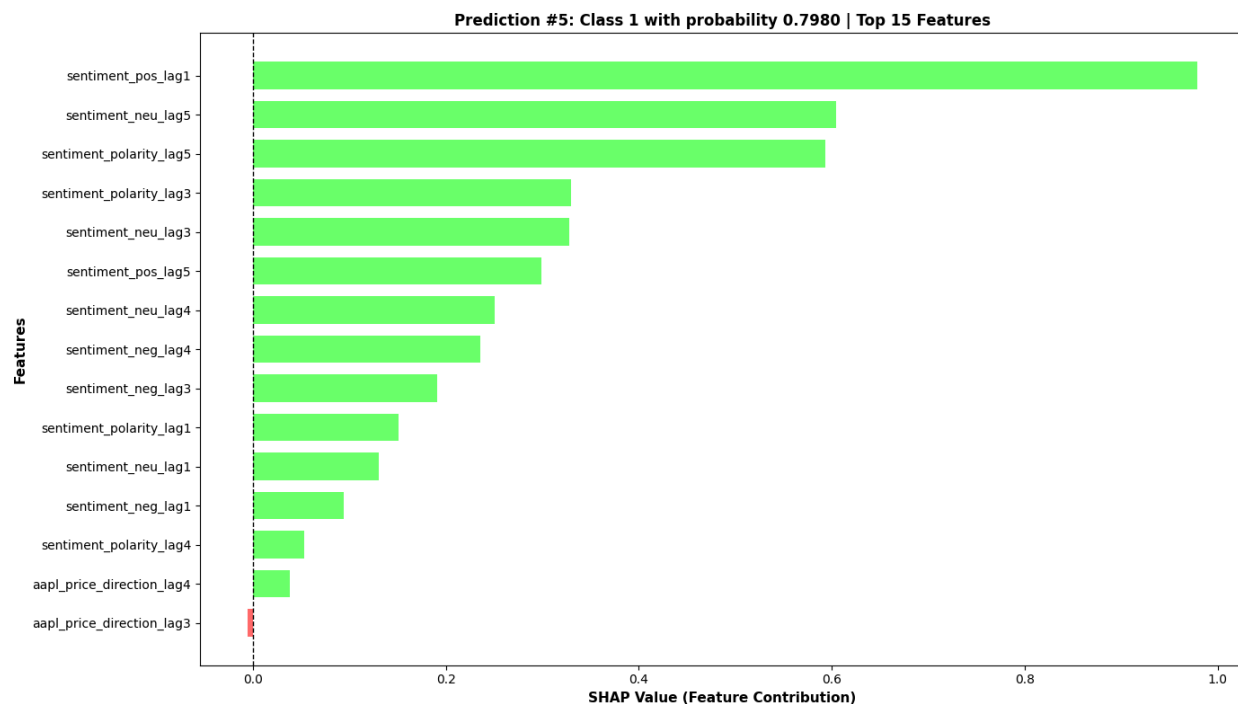
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PREDICTION #4 GRAPH

For Prediction #4, the model predicted Down with probability 0.3705, and this was correct, though less confidently than Prediction #3. This case is interesting because the model saw strong conflicting signals. Features such as sentiment_neg_lag2 (+1.0810), sentiment_pos_lag4 (+0.9011), sentiment_pos_lag1 (+0.6109), and sentiment_neg_lag3 (+0.4577) strongly supported an Up prediction. However, those were offset by several powerful downward features: sentiment_neu_lag5 (-0.9563), sentiment_neu_lag1 (-0.6353), sentiment_pos_lag3 (-0.6029), sentiment_neu_lag2 (-0.3889), and sentiment_polarity_lag2 (-0.3133). This means the final Down decision came from a tug-of-war between bullish and bearish sentiment lags, with the bearish side winning narrowly. To flip this case to Up, the most useful changes would be increasing sentiment_neu_lag5 from 0.8289 to 0.8792 and sentiment_neu_lag1 from 0.8361 to 0.8794, changes of 0.0503 and 0.0433, respectively. sentiment_neu_lag2 would only need to increase by 0.0114, and sentiment_polarity_lag2 would need a small increase of 0.0070. Since this prediction is closer to the midpoint than Prediction #3, it is more sensitive and likely easier to reverse.

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PREDICTION #4 GRAPH

For Prediction #5, the model predicted Up with probability 0.7980, and this was correct. The strongest features supporting Up were sentiment_pos_lag1 (+0.9790), sentiment_neu_lag5 (+0.6047), sentiment_polarity_lag5 (+0.5937), sentiment_polarity_lag3 (+0.3301), and sentiment_neu_lag3 (+0.3283). Several features did push toward Down, especially sentiment_polarity_lag2 (-0.4905), sentiment_neu_lag2 (-0.4459), sentiment_pos_lag2 (-0.3782), and sentiment_neg_lag5 (-0.3670), but the positive forces were larger overall. To flip this prediction to Down, the biggest adjustments would need to reduce the strongest bullish features. For example, sentiment_pos_lag1 would need to decrease from 0.1133 to 0.0826, a drop of 0.0308, while sentiment_neu_lag5 would need to decline by 0.0346. The largest required shifts are in polarity: sentiment_polarity_lag5 would need to drop from 0.7185 to 0.3863, a change of 0.3322, and sentiment_polarity_lag3 would need to fall from 0.7970 to 0.3820, a change of 0.4150. These are relatively large moves, which helps explain why the model was fairly confident in predicting Up.

Overall, these five examples show that the model is driven primarily by lagged sentiment features rather than lagged price direction, and especially by the most recent positive sentiment, second-day negative sentiment, fifth-day neutral sentiment, and fifth-day polarity. Features do not act in isolation; instead, predictions are created by the balance of several forces pushing toward Up or Down. Cases with probabilities near 0.50, like Prediction #1, can be flipped with moderate changes in a few important features, while more confident predictions such as #2, #3, and #5 would require larger and more coordinated shifts. The SHAP results therefore show not only which features matter most, but also that recent sentiment structure is the key mechanism through which the model makes its decisions.

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This dataset does not contain any protected categories. Our dataset does not contain any user information, and thus none of the features are present in the protected group.

There are many potential areas of bias with this model. The first is with the source of the news sentiment data. As we know firsthand, news in today's world can be manipulated to evoke emotion from the reader. Different news outlets may do this in different ways, and some report on facts more dependably than others. Our news sentiment dataset relies completely on Yahoo News, which is an aggregator of news. These articles are not fact-checked, and are collected by a variety of sources across the internet. A thorough bias correction would investigate the sources that supply news to Yahoo, but this is unfortunately impossible due to the omission of original article source in the obtained dataset.

Another possible source of bias with the news sentiment is the skew towards negativity. News organizations know that sensationalized headlines will often draw more readers, so they have a tendency to over-report on the negative news and under-report on the positive news. As a result, there is potentially more negatively categorized articles in the news sentiment dataset than is actually true under a neutral lens. It is also not surprising to see that articles with extreme negative sentiment language have a high impact on stock price direction, as these consumers are likely readers that may react to negative mentions in the news.

The final source of bias is Apple news being categorized not just because of their own company news, but because of their business competition or other companies in the writer's portfolio. This is especially evident when Apple is being compared to other large technology companies, such as Amazon or Tesla. The news article will comment on "large tech companies" stock performance, and will tag Apple in the keywords, but it doesn't actually refer to the individual company when speaking in generalities. In other words, Apple's news sentiment is biased by group association. Below is another example:

"S&P 500 ends at new record on stellar tech earnings, surge in U.S. economic growth recorded modest gains on Thursday after a round of upbeat earnings reports from tech heavyweights. Saw some turbulence midday as Treasury yields rose and investors questioned how much good news was priced into the market."

Apple is biased here in their own news because they are being associated with the performance of the rest of the S&P 500, not because of their own stock. We account for that in the extra-credit assignment by retraining a model on data that **only explicitly mentions the Apple stock in title or content**. Below is the performance comparison with Week 10's model:

	Accuracy		Macro Average F1 Score	
	Validation Dataset	Test Dataset	Validation Dataset	Test Dataset
Week 11 Model	55%	53%	0.55	0.53
Week 10 Model	70%	52%	0.70	0.46

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This week's model with reduced bias actually surpassed Week 10's model on the test dataset in both accuracy and F1 score, despite achieving lower performance on the validation set. This suggests that the bias removal helped with overfitting, which was a large problem in previous weeks, and removed some of the "noise" that previously caused this model to train on false patterns in the data. In addition, the F1 scores for the "increase" and "decrease" classes were only separated by 0.01, a balance no prior model had been able to achieve.

Stakeholder Risks

If our model were to be presented to stakeholders, there are a few potential risks that would be important to keep in mind.

First, we would need to ensure that stakeholders do not over-rely on our model for making financial decisions. The stock market is notoriously difficult to predict, and while the model had modest positive results, it is far from perfect at predicting which way the market will move. In a presentation to stakeholders, we would need to stress that our model is only one element of a complete investment strategy.

We should also consider that our final model, XGBoost, can be difficult to interpret. XGBoost models consist of an ensemble of many decision trees that capture complex, non-linear relationships. This structure makes it difficult to determine how individual features impact predictions. While our model's performance may be easy to communicate with stakeholders, it would be considerably more difficult to communicate how our model comes up with its predictions, especially if the stakeholders are unfamiliar with tree-based models. It would require careful visualizations and attention to detail in order to adequately help our stakeholders interpret our model.